MetaEvoMining: a tool for the exploration of novel biosynthetic pathways applied to polluted ocean metagenomic data

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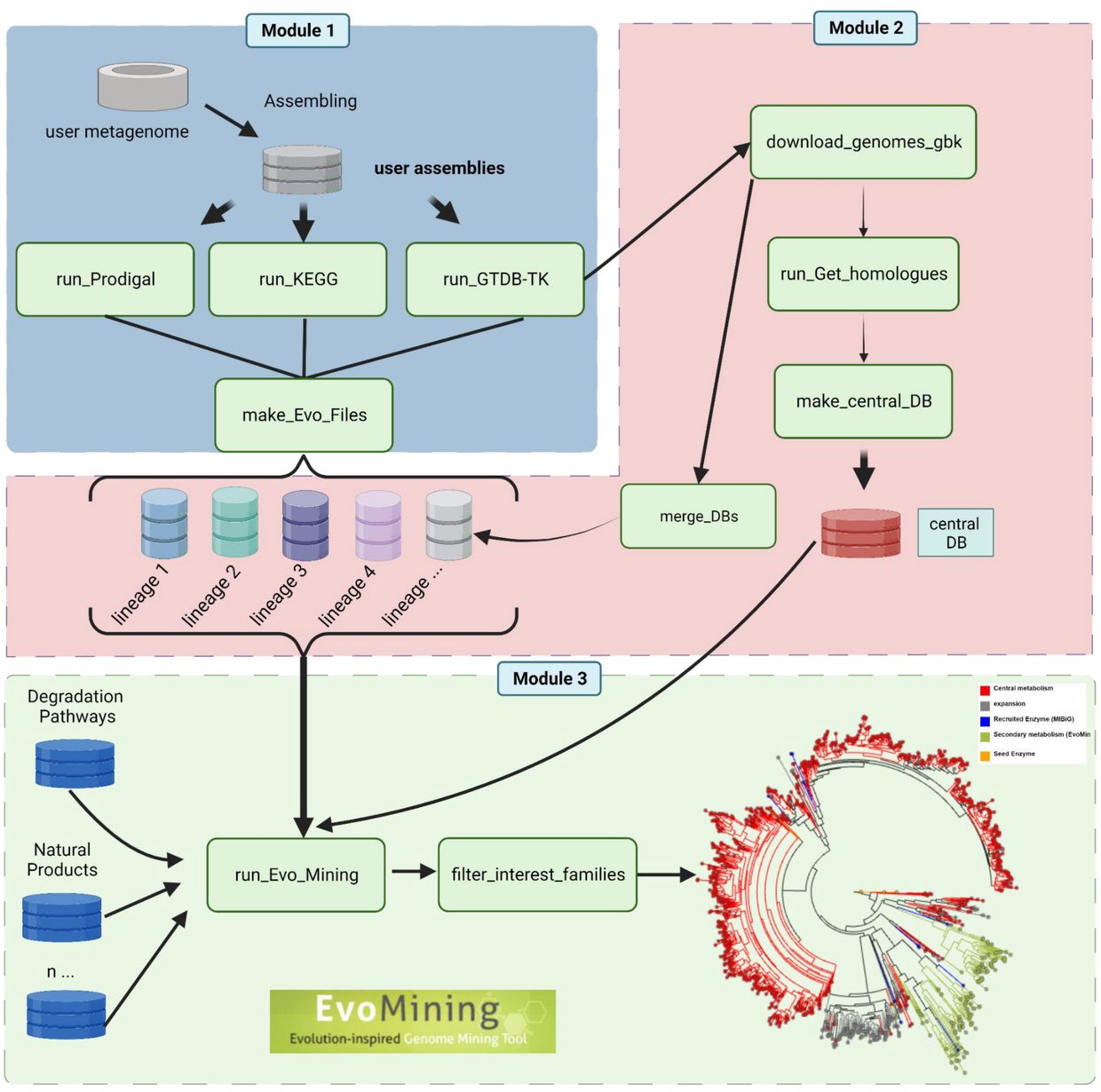
Abstract

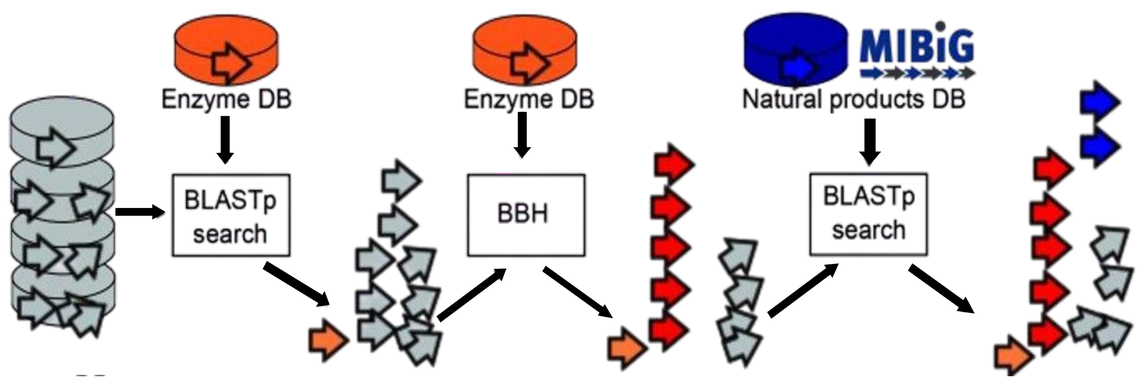
The collection of metagenomic data has increased in the last 20 years. In genomic data, EvoMining is a tool that has been used as a strategy to discover new enzymes recruited by the specialized metabolism, but has not been used in metagenomic data yet.

To connect EvoMining with metagenomic data, we are developing an open-source package: MetaEvoMining is a tool that allows us to mine metagenomic and genomic data and find useful knowledge about new metabolic pathways through phylogenies.

The pipeline of the MetaEvoMining package has three modules.

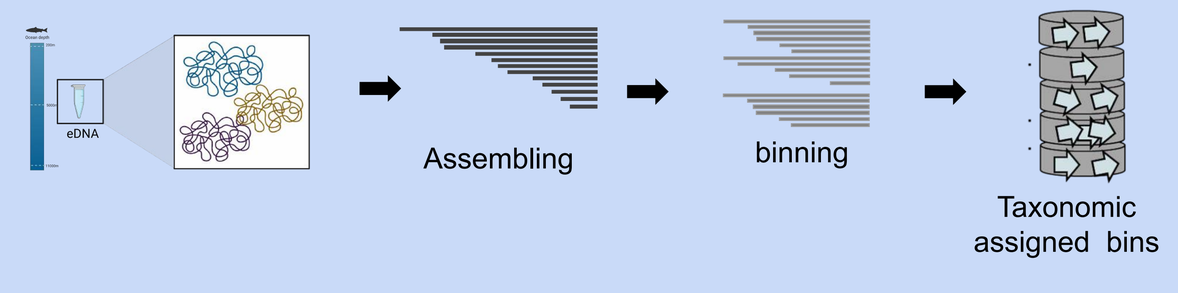


EvoMining find expansions and recruitment of enzymes in taxonomically related genomes.

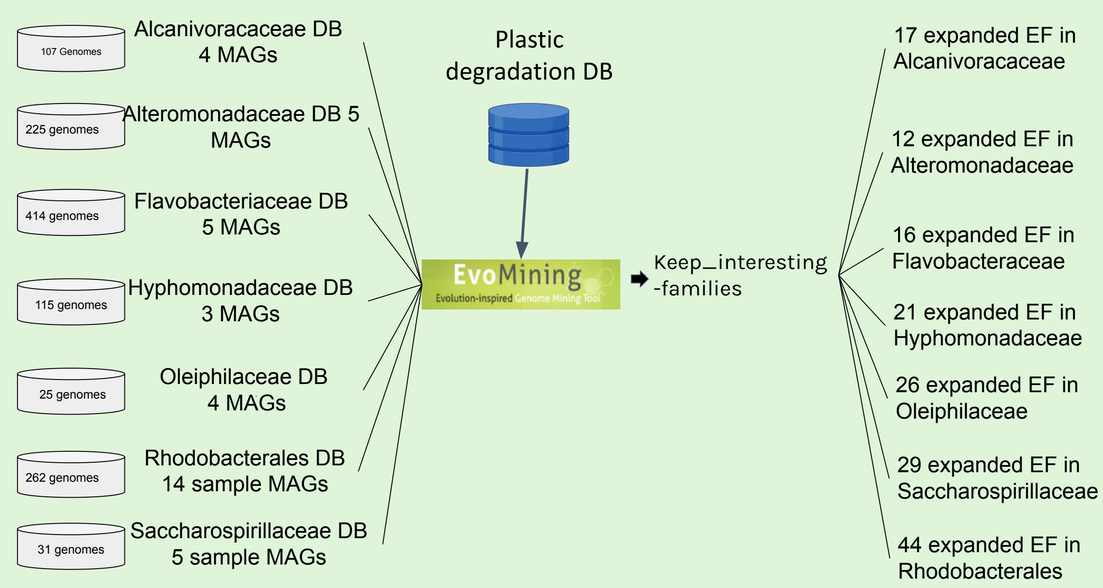


EvoMining uses the reconstructed genomic information for mining into metagenomes by sequence homology to find candidates for new key enzymes.

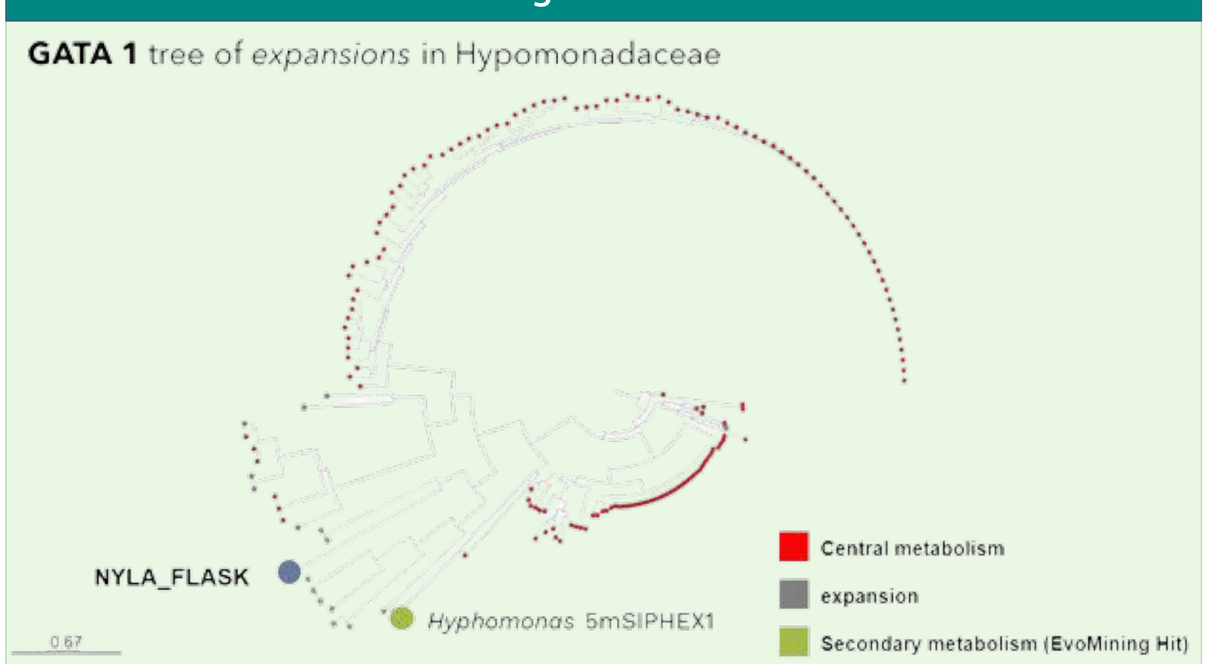
Metagenomics can lead to discovering new biosynthetic pathways.



MetaEvoMining found expansions in PET degradation enzymes



The prediction of recruitment to a new pathway for plastic degradation.



Conclusions and perspectives.

Through our package, we were able to identify gene families that play a role in new metabolic pathways. Our team developed a reproducible and automatic pipeline using a single R package to analyze a metagenomic sample. We were able to identify enzymes that have potential biochemical capabilities not previously described by functional annotation. Additionally, we found expanded families that could be promising candidates for new chemistry. The expansion we discovered in the enzymatic families could provide valuable insight into the new routes microorganisms are taking to degrade new substrates.

References:

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